

CS & IT ENGINEERING

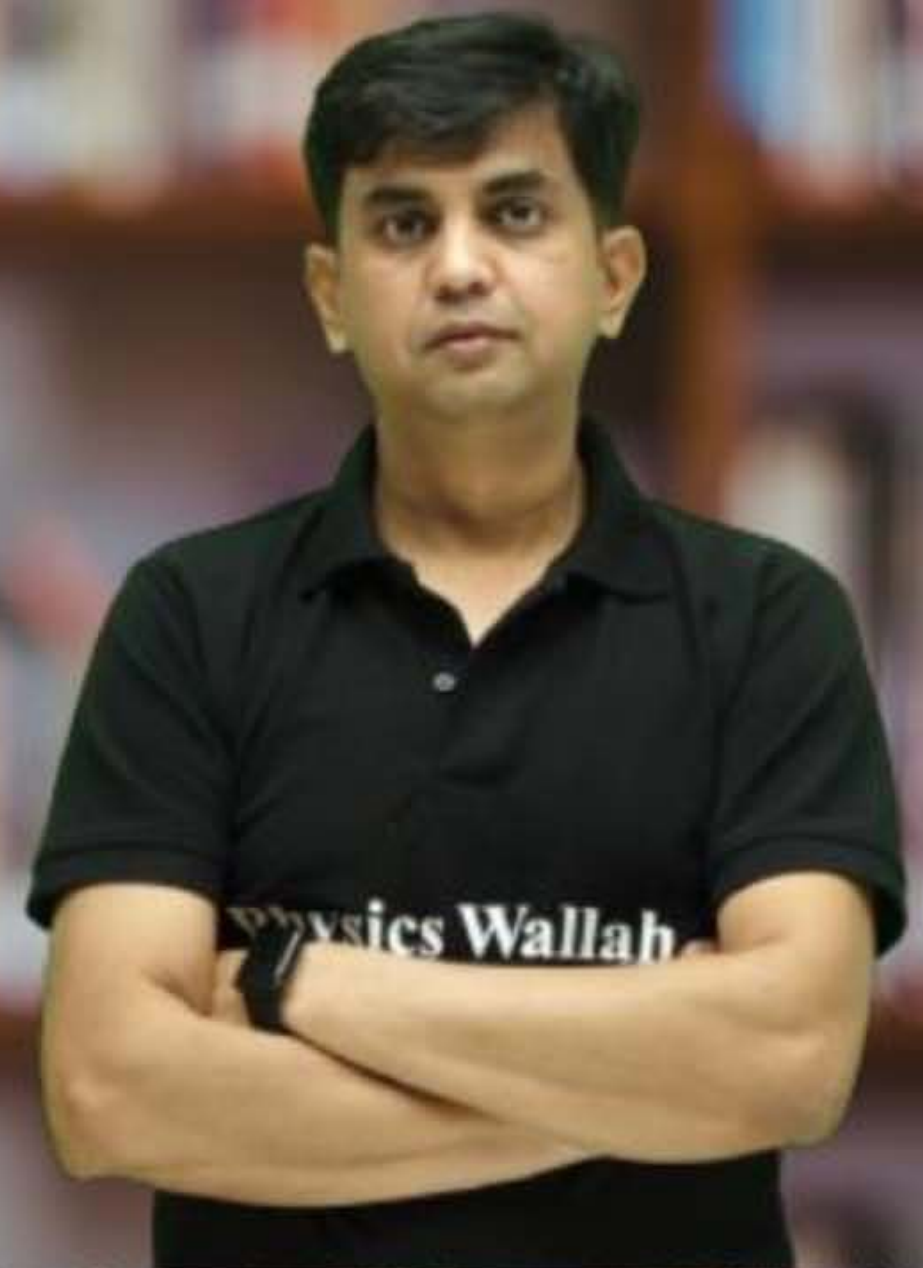


Computer Network

Flow Control

Lecture No. - 08

By - Abhishek Sir





Recap of Previous Lecture



Topic

Go Back N ARQ

Topic

Selective Repeat ARQ



Topics to be Covered



Topic

Selective Repeat ARQ

Topic

End-to-End Delay

ABOUT ME



Hello, I'm **Abhishek**

- GATE CS AIR - 96
- M.Tech (CS) - IIT Kharagpur
- 12 years of GATE CS teaching experience

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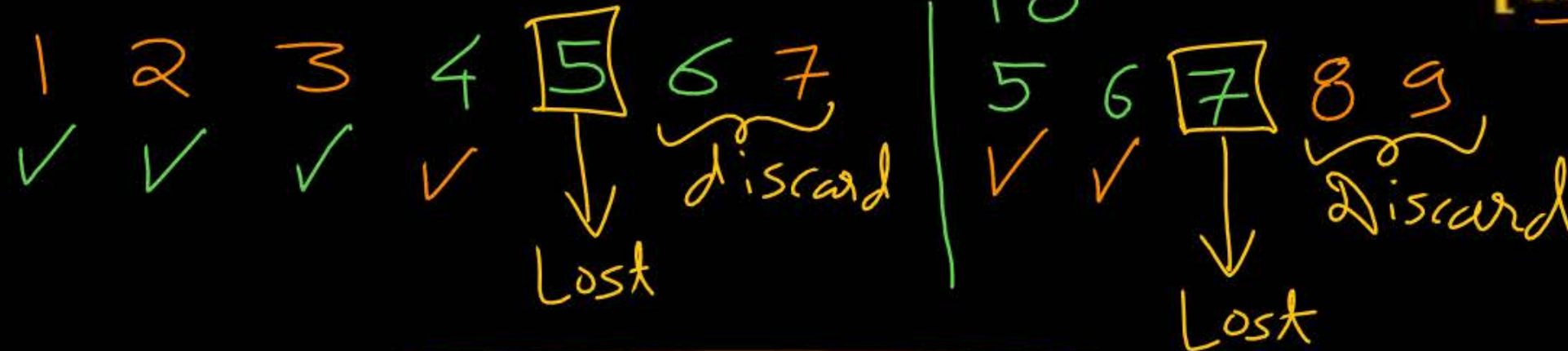


#Q. Station A needs to send a message consisting of 9 packets to Station B using a sliding window (window size 3) and go-back-n flow control strategy. All packets are ready and immediately available for transmission. If every 5th packet that A transmits gets lost (but no ACKs from B ever get lost), then what is the number of packets that A will transmit for sending the message to B?

[GATE 2006]

A

12

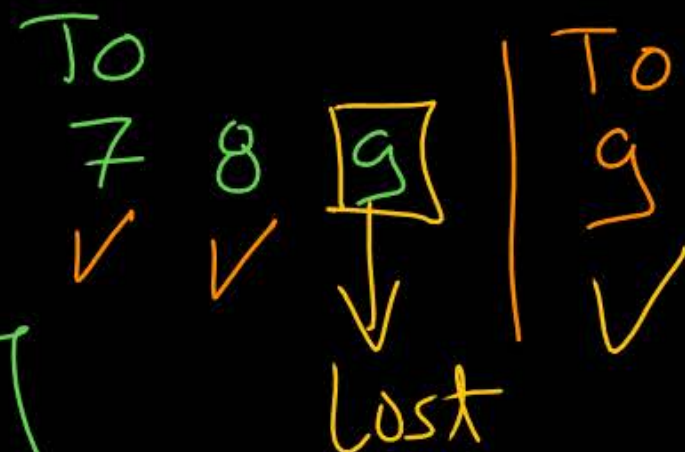


B

14

C

16



D

18

Ans: C



Transmitter

Receiver

Transmitter

Receiver

$F_1 F_2 F_3$

F_1
 F_2
 F_3

$F_1 \checkmark$
 $F_2 \checkmark$
 $F_3 \checkmark$

A_1
 A_2
 A_3

$F_4 F_5 F_6$

F_4
 F_5
 F_6

$F_4 \checkmark$
 F_6 (discard)
 F_5 (discard)

F_4
 F_5
 F_6

A_4
 A_4
 A_4

F_7

ignore $\leftarrow$

F_7

A_4

F_7 (discard)

TO F_5

F_5
 F_6
 F_7

$F_5 \checkmark$
 $F_6 \checkmark$

A_5
 A_6

F_8
 F_9

F_8
 F_9

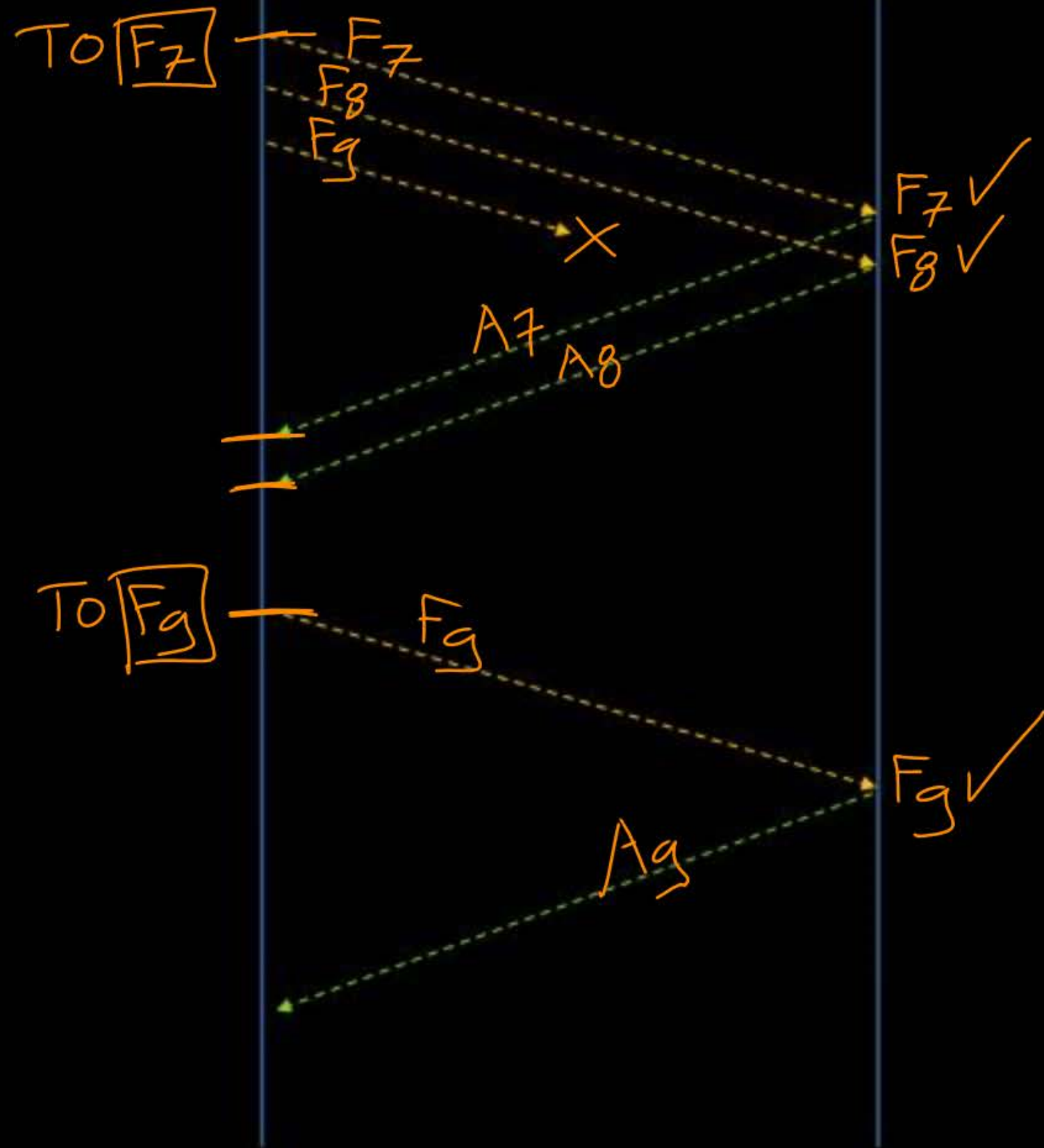
F_8 (discard)
 F_9 (—/—)

A_6
 A_6

ignore $\leftarrow$

Transmitter

Receiver



#Q. Consider a network connecting two systems located 8000 kilometers apart. The bandwidth of the network is 500×10^6 bits per second. The propagation speed of the media is 4×10^6 meters per second. It is needed to design a Go-Back-N sliding window protocol for this network. The average packet size is 10^7 bits. The network is to be used to its full capacity. Assume that processing delays at nodes are negligible. Then, the minimum size in bits of the sequence number field has to be _____.

[GATE 2015, Set-3, 2-Mark]

Solution:-

$$\text{Packet Size} = 10^7 \text{ bits}$$

$$\text{Bandwidth} = 500 * 10^6 \text{ bits / sec}$$

$$t_x = \frac{\text{Packet Size}}{\text{Bandwidth}} = \frac{10^7 \text{ bits}}{500 * 10^6 \text{ bits / sec}} = \frac{1}{50} \text{ sec}$$

$$\text{Distance} = 8000 \text{ Km} = 8 * 10^6 \text{ m}$$

$$\text{Signal Speed} = 4 * 10^6 \text{ m/s}$$

$$t_p = \frac{\text{Distance}}{\text{Signal Speed}} = \frac{8 * 10^6 \text{ m}}{4 * 10^6 \text{ m/s}} = 2 \text{ sec}$$

$$\text{Cycle Time} = (t_x + 2 * t_p) = \left(\frac{1}{50} + 2 * 2 \right) \text{ sec} = \frac{201}{50} \text{ sec}$$

$$\text{Optimal Window Size (N)} = \left\lceil \frac{\text{Cycle Time (RTT)}}{\text{Transmission delay}} \right\rceil = 201$$

For Go Back N ARQ :

$$\text{Total number of sequences} = (N + 1) = 202$$

Minimum number of bits required for sequence number field

$$= \left\lceil \log_2 [\text{Total number of sequences}] \right\rceil \text{ bits}$$

$$= \left\lceil \log_2 (202) \right\rceil \text{ bits}$$

$$= 8 \text{ bits}$$

$$\boxed{\text{Ans} = 8}$$

#Q. A 20 Kbps satellite link has a propagation delay of 400 ms. The transmitter employs the "go back n ARQ" scheme with n set to 10. Assuming that each frame is 100 bytes long, what is the maximum data rate possible?

Throughput

[GATE 2004]

(A) 5Kbps

✓ (B) 10Kbps

(C) 15Kbps

(D) 20Kbps

Ans: B

Solution:-

$$\text{Frame Size} = 100 \text{ bytes} = 8 * 10^2 \text{ bits}$$

$$\text{Bandwidth} = 20 \text{ Kbps} = 2 * 10^4 \text{ bits / sec}$$

$$t_x = \frac{\text{Frame Size}}{\text{Bandwidth}} = \frac{8 * 10^2 \text{ bits}}{2 * 10^4 \text{ bits / sec}} = 40 \text{ ms}$$

$$t_p = 400 \text{ ms}$$

$$\begin{aligned} \text{Cycle time} &= (t_x + 2 * t_p) = \cancel{840 \mu s} \quad 840 \text{ ms} \\ &= (40 + 2 * 400) \text{ ms} \end{aligned}$$

$$\text{Window Size} = 10$$

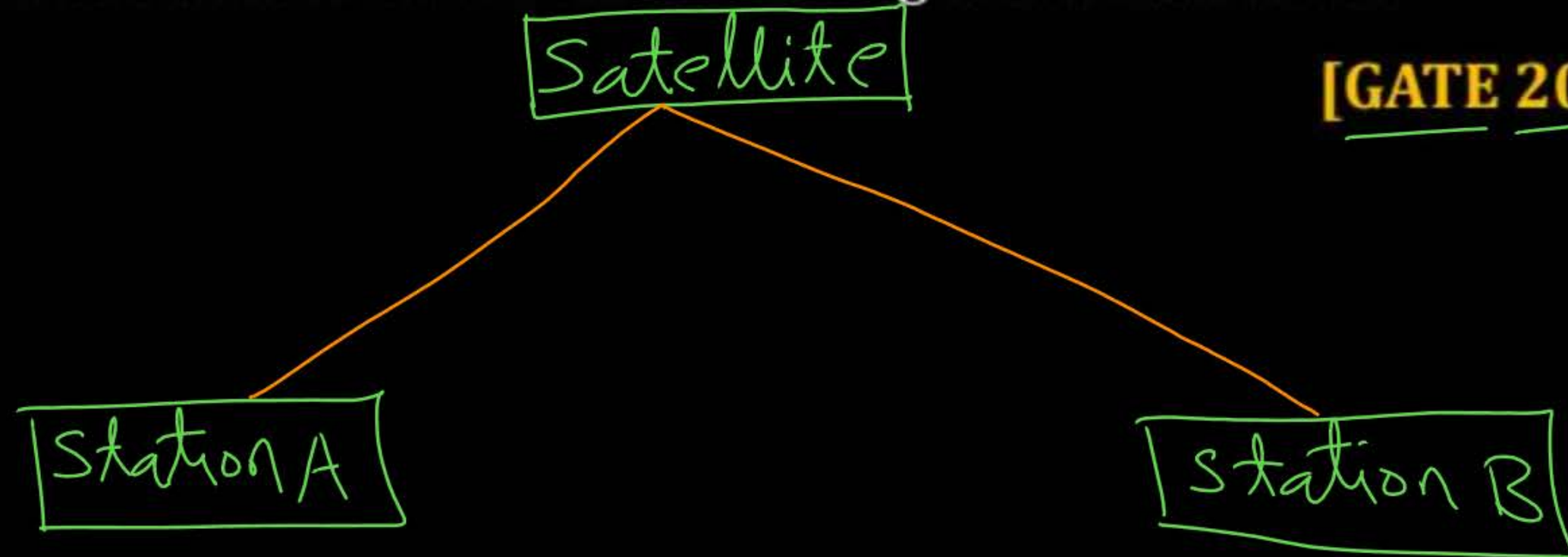
$$\text{Throughput} = \frac{\text{Window Size} * \text{Packet Size}}{\text{Cycle Time}}$$

$$= \frac{10 * 800 \text{ bits}}{840 \text{ ms}} = \frac{8000}{840} * 10^3 \text{ bits/sec}$$

$$= \frac{800}{84} \text{ kbps} \approx 10 \text{ Kbps}$$

#Q. A 1Mbps satellite link connects two ground stations. The altitude of the satellite is 36,504 km and speed of the signal is 3×10^8 m/s. What should be the packet size for a channel utilization of 25% for a satellite link using go-back-127 sliding window protocol? Assume that the acknowledgment packets are negligible in size and that there are no errors during communication.

[GATE 2008]



- ✓ (A) 120 bytes
- (B) 60 bytes
- (C) 240 bytes
- (D) 90 bytes

Ans: A

Solution:-

Packet Size = ? (in bytes)

Bandwidth = 1 Mbps = 10^6 bits / sec

Distance = $2 * 36504$ Km = $73008 * 10^3$ m

Signal Speed = $3 * 10^8$ m/s

$$t_p = \frac{\text{Distance}}{\text{Signal Speed}} = \frac{[73008 * 10^3 \text{ m}]}{[3 * 10^8 \text{ m/s}]} = [24336 * 10^{-5} \text{ Sec}]$$

To achieve 25% utilization ($\eta = 1/4$) in Go Back 127 ARQ

$$\text{Efficiency } (\eta) = \frac{N * \text{Transmission delay}}{\text{Cycle Time}}$$

$$\text{Cycle Time} = 4 * N * \text{Transmission delay}$$

$$(t_x + 2 * t_p) = 4 * 127 * t_x$$

$$t_x = 2 * t_p / 507$$

$$\text{Packet Size} = (2 * t_p / 507) * \text{Bandwidth}$$

$$= (2 * 24336 * 10^{-5} \text{ Sec} / 507) * 10^6 \text{ bits / sec}$$

$$= 960 \text{ bits} = 120 \text{ bytes}$$



Topic : Selective Repeat ARQ

→ Transmitter's transmitting window size = N

→ Receiver's receiving window size = N

$(N > 1)$

→ Total number of sequences = $2N$ [0 to $(2N-1)$]

Total number of sequences =
Transmitter's transmitting window size
+ Receiver's receiving window size

$\Rightarrow$ Alternet sequence set protocol.

Sequence number $\leftarrow$ (Frame number) mod $(2N)$



Topic : Selective Repeat ARQ



CASE I :

Suppose

$N = 4$

Sequence Number = 0 to 7

[0-3]

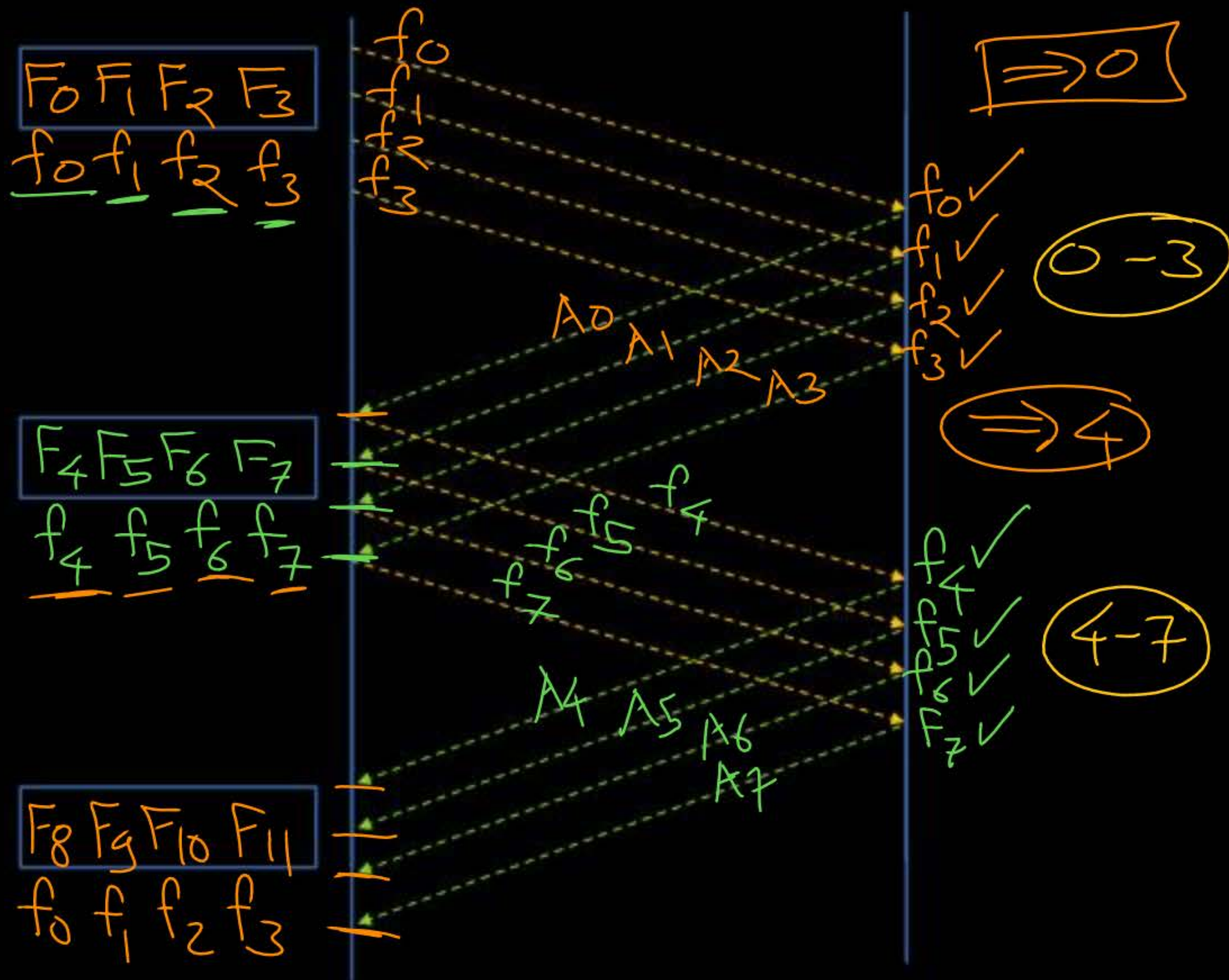
[4-7]

[0-3]

[4-7]

Transmitter

Receiver





Topic : Selective Repeat ARQ



CASE II :

Suppose

$N = 4$

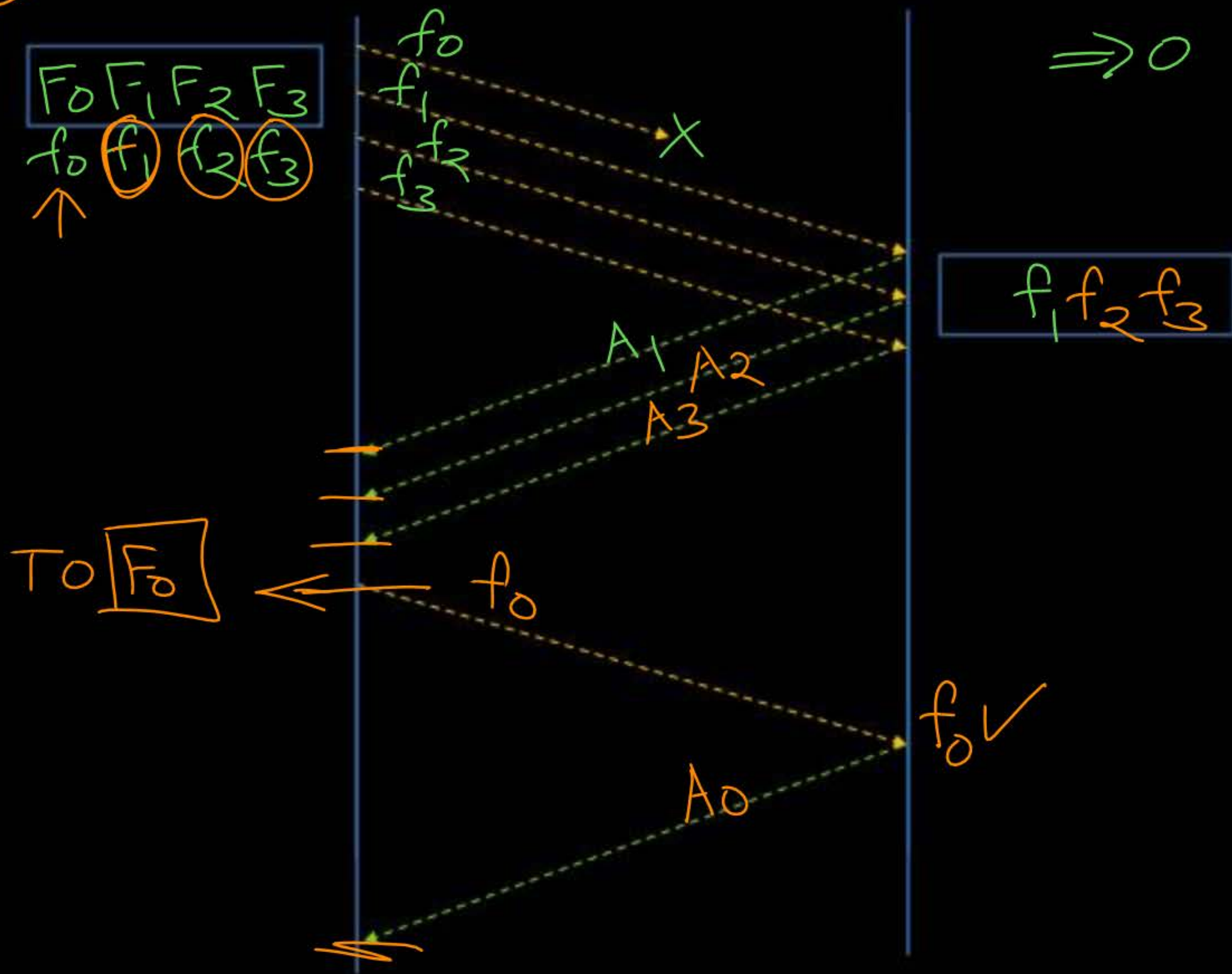
Sequence Number = 0 to 7

Selectively
Reject ARQ

$F_0 F_1 F_2 F_3$
 $f_0 f_1 f_2 f_3$
↑

Transmitter

Receiver





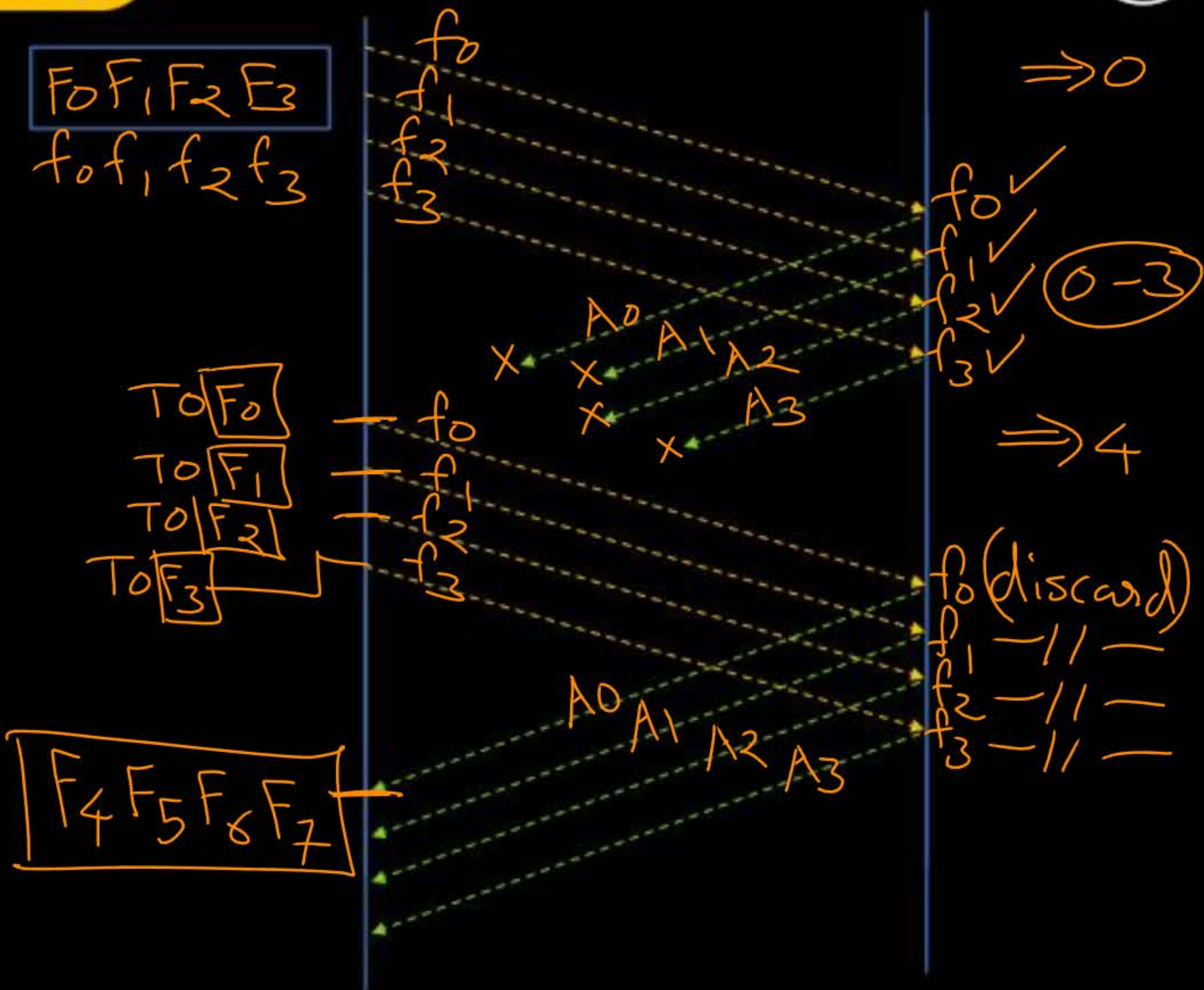
Topic : Selective Repeat ARQ



CASE III :

Suppose $N = 4$

Sequence Number = 0 to 7





Topic : Selective Repeat ARQ



- Transmitter transmit **N** frames **without any acknowledgment**
- Receiver transmit “**individual acknowledgment**”
[for every successfully received frame]
- “Cumulative (combine) acknowledgment” does not exist in this protocol



Topic : Selective Repeat ARQ



→ Whenever transmitter gets time-out or received NACK, it retransmit that particular frame only [mostly first frame, resides in transmitting window]

→ Receiver buffer the frame which is out of order (expected sequence numbers) and send individual acknowledgment of that frame

Expected seq. no. range
= Expected seq. no., Expected seq. no. + N - 1

* Selective Reject ARQ

#Q. Consider a 128×10^3 bits/second satellite communication link with one-way propagation delay of 150 milliseconds. Selective retransmission (repeat) protocol is used on this link to send data with a frame size of 1 kilobyte. Neglect the transmission time of acknowledgment. The minimum number of bits required for the sequence number field to achieve 100% utilization is _____.

[GATE-2016, Set-2, 2-Mark]

11SC / H.W.

#Q. Consider a selective repeat sliding window protocol that uses a frame size of 1 KB to send data on a 1.5 Mbps link with a one-way latency of 50 msec. To achieve a link utilization of 60%, the minimum number of bits required to represent the sequence number field is _____.

[GATE-2014, Set-1, 2-Mark]

IIT-KGP, H.W.

#Q. In a sliding window ARQ scheme, the transmitter's window size is N and the receiver's window size is M . The minimum number of distinct sequence numbers required to ensure correct operation of the ARQ scheme is

[GATE 2004]

IIT-D, H.W.

- A** $\min(M, N)$
- B** $\max(M, N)$
- C** $M + N$
- D** MN



2 mins Summary



Topic

Selective Repeat ARQ ✓

Topic

~~End-to-End Delay~~



THANK - YOU

